Electricity & Magnetism

One-Shot Revision Notes (Serial Wise)

Exam Revision Version — High Yield Notes

Contains formulas, definitions, crucial derivations, TikZ diagrams, and high-weightage topics curated directly from PYQs.

Designed for Quick Revision & Exam Readiness

May 31, 2026

Contents

1 Part 1: Electrostatics

Important Note: This is the condensed exam-oriented revision version. Focus heavily on memorizing the boxed formulas, definitions, and high-weightage derivations listed under the PYQ sections.

1.1 Electric Charge

Electric charge is the fundamental intrinsic property of matter that gives rise to electric forces.

1.1.1 Key Properties

1. **Quantization of Charge:** Charge exists only in discrete integral multiples of the elementary charge e.

   Quantization of Charge

   $$q = ne$$

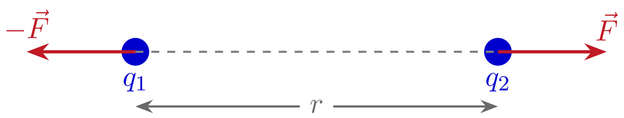
   Where $e = 1.6 \times 10^{-19}$ C is the charge of a proton/electron, and $n \in \mathbb{Z}$.

2. **Conservation of Charge:** The total net electric charge of an isolated system remains constant over time.

3. **Additivity of Charge:** Total charge of a system is the algebraic sum of all individual charges: $q_{\text{total}} = q_1 + q_2 + q_3 + \dots$

1.2 Coulomb's Law

Provides the electrostatic force of attraction or repulsion between two stationary point charges.



Coulomb's Law Formula

$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}$$

Where ϵ_0 is the permittivity of free space ($\epsilon_0 = 8.854 \times 10^{-12}\ \mathrm{C^2/N \cdot m^2}$).

Electrostatic Constant

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \mathrm{N \cdot m^2/C^2}$$

1.3 Electric Field Intensity

The electric field intensity $\vec{E}$ at any point is defined as the electrostatic force experienced per unit positive test charge placed at that point.

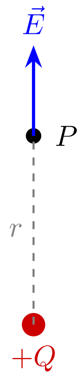
Electric Field Strength

$$E = \frac{F}{q} \implies \vec{E} = \frac{\vec{F}}{q}$$

Unit: N/C (or V/m)

1.4 Electric Field Due to a Point Charge

Determines the electric field produced by a single source charge $+Q$ at a distance r.



Electric Field (Point Charge)

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

1.5 Superposition Principle

The net electric field at any point due to a system of multiple stationary charges is the vector sum of the electric fields produced by each individual charge at that point.

Superposition Principle

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \cdots = \sum_{i=1}^{n} \vec{E}_i$$

1.6 Electric Flux

Electric flux is a measure of the number of electric field lines passing through a given surface area A.

Electric Flux Formula

$$\phi_E = \vec{E} \cdot \vec{A} = EA\cos\theta$$

Where θ is the angle between the electric field vector $\vec{E}$ and the normal vector to the surface area $\vec{A}$.

Unit: $\text{N} \cdot \text{m}^2/\text{C}$

1.6.1 Special Cases

- **Maximum Flux ($\theta = 0°$):** The field lines are normal to the surface.

$$\phi_E = EA$$

- **Zero Flux ($\theta = 90°$):** The field lines run parallel to the surface.

$$\phi_E = 0$$

1.7 Gauss's Theorem

PYQ Priority: *****

Statement: The net electric flux ϕ_E through any closed Gaussian surface is equal to $\frac{1}{\epsilon_0}$ times the net charge enclosed within that surface.

Gauss's Law (Integral Form)

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Gauss's Law (Differential Form)

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

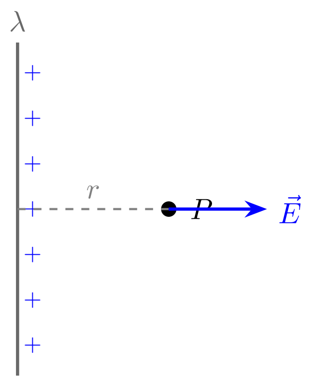
Where ρ is the volume charge density.

1.8 Applications of Gauss's Law

1.8.1 (A) Infinite Line Charge

PYQ Priority: *****

Calculates the electric field at a perpendicular distance r from an infinitely long wire with uniform linear charge density λ.



Electric Field of Infinite Line Charge

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

1.8.2 (B) Infinite Plane Sheet

PYQ Priority: *****

Calculates the electric field produced by an infinite plane sheet of charge with uniform surface charge density σ.

Electric Field of Infinite Plane Sheet

$$E = \frac{\sigma}{2\epsilon_0}$$

Crucial Note: The electric field is completely **independent of the distance** r from the sheet.

1.8.3 (C) Uniformly Charged Sphere

For a solid insulating sphere of radius R and total uniform charge Q:

- **Outside the Sphere** ($r \geq R$):

  Outside Sphere

  $$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

- **Inside the Sphere** ($r < R$):

  Inside Sphere

  $$E = \frac{1}{4\pi\epsilon_0}\frac{Qr}{R^3}$$

1.9 Electric Potential

The electric potential V at any point in an electric field is the amount of work done per unit positive charge in bringing it from infinity to that point against electrostatic forces.

Electric Potential Formula

$$V = \frac{W}{q}$$

Unit: Volt (V) = J/C

1.10 Potential Due to a Point Charge

Potential (Point Charge)

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}$$

1.11 Relation Between Electric Field and Potential

PYQ Priority: *****

Defines the relationship between the electric field intensity and the electric potential gradient.

Relation Between E and V

$$E = -\frac{dV}{dr}$$

Vector Form (Gradient):

$$\vec{E} = -\nabla V$$

1.12 Potential Energy

The electrostatic potential energy U of a charge q in an electric potential V is:

Potential Energy (Single Charge)

$$U = qV$$

For a system of two point charges q_1 and q_2 separated by a distance r:

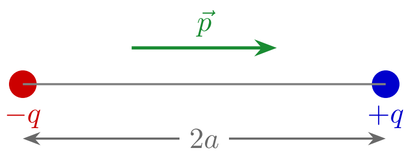
Potential Energy (Two Charges)

$$U = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r}$$

1.13 Electric Dipole

PYQ Priority: *** (Direct 10 Marks Derivation)**

A pair of equal and opposite point charges $+q$ and $-q$ separated by a very small distance $2a$.



Dipole Moment

$$p = q(2a) \quad \Longrightarrow \quad \vec{p} = q(2\vec{a})$$

The vector direction of $\vec{p}$ is always from the **negative charge to the positive charge**.
Unit: $\mathrm{C \cdot m}$

1.14 Potential Due to an Electric Dipole

PYQ Priority: *****

General formula for the potential V at any position (r, θ) relative to the dipole center.

Potential of a Dipole (General)

$$V = \frac{1}{4\pi\epsilon_0}\frac{p\cos\theta}{r^2}$$

- **Axial Position** ($\theta = 0°$):

$$V_{\text{axial}} = \frac{1}{4\pi\epsilon_0}\frac{p}{r^2}$$

- **Equatorial Position** ($\theta = 90°$)**:**

$$V_{\text{equatorial}} = 0$$

1.15 Electric Field Due to a Dipole

- **Axial Field (End-on Position):**

Axial Electric Field

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0}\frac{2p}{r^3}$$

- **Equatorial Field (Broadside-on Position):**

Equatorial Electric Field

$$E_{\text{equatorial}} = \frac{1}{4\pi\epsilon_0}\frac{p}{r^3}$$

- **General Position Field:**

General Electric Field

$$E = \frac{1}{4\pi\epsilon_0}\frac{p}{r^3}\sqrt{1 + 3\cos^2\theta}$$

1.16 Torque on an Electric Dipole

When placed in a uniform external electric field $\vec{E}$:

Torque on Dipole

$$\tau = pE\sin\theta$$

Vector Form:

$$\vec{\tau} = \vec{p} \times \vec{E}$$

1.17 Potential Energy of a Dipole

The potential energy stored in a dipole placed in a uniform electric field $\vec{E}$:

Potential Energy of Dipole

$$U = -pE\cos\theta = -\vec{p}\cdot\vec{E}$$

1.18 Polarization

The polarization $\vec{P}$ is defined as the induced electric dipole moment per unit volume of a dielectric material.

Polarization

$$P = \frac{\text{Electric Dipole Moment}}{\text{Volume}}$$

Unit: $\mathrm{C/m^2}$

1.19 Relation Between Polarization and Surface Charge Density

PYQ Priority: ★★★★

Relates the polarization vector to the induced surface bound charge density σ_p.

Polarization and Bound Charge

$$\sigma_p = P$$

General Vector Form:

$$\sigma_p = \vec{P} \cdot \hat{n}$$

Where $\hat{n}$ is the unit normal vector pointing outward from the dielectric surface.

1.20 Dielectric Constant

The ratio of permittivity of the medium to the permittivity of free space.

Dielectric Constant

$$K = \frac{\epsilon}{\epsilon_0}$$

1.21 Electric Susceptibility

Measures how easily a dielectric polarizes when subjected to an external electric field.

Electric Susceptibility

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

Where χ_e is the dimensionless electric susceptibility.

1.21.1 Key Relation

Relation Between K and Susceptibility

$$K = 1 + \chi_e$$

1.22 Electric Displacement Vector

The electric displacement vector $\vec{D}$ accounts for the effects of both free and bound charges in a dielectric material.

Electric Displacement Vector

$$\vec{D} = \epsilon \vec{E}$$

Alternative Form:

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

1.23 Gauss's Law in Dielectrics

In a dielectric medium, the total flux of the electric displacement vector $\vec{D}$ through any closed surface is equal to the net free charge enclosed within that surface. Bound charges are implicitly accounted for by the polarization term inside $\vec{D}$.

Gauss's Law in Dielectrics

$$\oint \vec{D} \cdot d\vec{A} = Q_{\text{free}}$$

Differential Form:

$$\nabla \cdot \vec{D} = \rho_{\text{free}}$$

1.24 Capacitor with Dielectric

When a dielectric material of dielectric constant K is introduced between the plates of a parallel plate capacitor, its capacitance always increases from C_0 to $C = KC_0$. However, other electrical quantities depend on whether the battery remains connected or is isolated.

1. **Case 1: Battery Remains Connected (Voltage Constant)**
   - **Potential Difference:** $V = V_0$ (remains constant).
   - **Capacitance:** $C = KC_0$ (increases).
   - **Charge:** $Q = KQ_0$ (increases as more charge is drawn from the battery).
   - **Electric Field:** $E = E_0$ (remains constant since V and d are constant).
   - **Energy Stored:** $U = \frac{1}{2}CV^2 = KU_0$ (increases).
2. **Case 2: Capacitor is Isolated after Charging (Charge Constant)**
   - **Charge:** $Q = Q_0$ (remains constant since no path exists for charge flow).
   - **Capacitance:** $C = KC_0$ (increases).
   - **Potential Difference:** $V = \frac{V_0}{K}$ (decreases by a factor of K).
   - **Electric Field:** $E = \frac{E_0}{K}$ (decreases due to induced bound charge fields).
   - **Energy Stored:** $U = \frac{Q^2}{2C} = \frac{U_0}{K}$ (decreases).

Electrostatics PYQ Priority: High-Yield Topics

Make sure you can thoroughly write and derive the following high-weightage topics:

1. **Gauss's Theorem:** Derivation and physical meaning.
2. **Infinite Line Charge Field:** Gaussian cylinder application.
3. **Electric Dipole potential and field** at axial, equatorial, and general positions.
4. **Polarization relation** ($\sigma_p = \vec{P} \cdot \hat{n}$).
5. **Dielectric Constant & Electric Displacement Vector** definitions.

Electrostatics Revision Complete! ✓

2 Part 2: Magnetostatics

2.1 Magnetic Field

The vector space around a moving charge, current-carrying conductor, or magnetic material where magnetic forces can be detected.

Magnetic Field Symbol and Units

Symbol: $\vec{B}$

Unit: Tesla (T) $= 1$ N/(A $\cdot$ m) $= 10^4$ Gauss

2.2 Magnetic Field Lines

Imaginary lines depicting the direction and density of the magnetic field vector.

2.2.1 Crucial Properties

- Form continuous **closed loops** (no magnetic monopoles).
- **Outside the magnet:** Point from North (N) to South (S).
- **Inside the magnet:** Point from South (S) to North (N).
- They never intersect.

2.3 Lorentz Force

PYQ Priority: ★★★★

Total force experienced by a charged particle moving through both electric and magnetic fields.

Lorentz Force (Combined)

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

2.3.1 Magnetic Force Only

When only a magnetic field is present:

Magnetic Lorentz Force

$$F = qvB\sin\theta$$

Where θ is the angle between the velocity vector $\vec{v}$ and the magnetic field vector $\vec{B}$.

Special Cases:

- **Parallel Motion ($\theta = 0°$ or $180°$):** $F = 0$. The path remains straight.
- **Perpendicular Motion ($\theta = 90°$):** $F = qvB$ (Maximum force). The path becomes circular.

2.4 Force on a Current-Carrying Conductor

Magnetic Force on Wire

$$F = BIL\sin\theta$$

Vector Form:

$$\vec{F} = I(\vec{L} \times \vec{B})$$

Where I is the current, $\vec{L}$ is the length vector in the direction of the current, and $\vec{B}$ is the magnetic field.

2.5 Motion of a Charged Particle in a Uniform Magnetic Field

For a particle entering perpendicularly ($\theta = 90°$):

- **Radius of the Circular Path:**

Circular Radius

$$r = \frac{mv}{qB}$$

- **Time Period of Revolution:**

Time Period

$$T = \frac{2\pi m}{qB}$$

- **Frequency of Revolution:**

Frequency

$$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

2.6 Biot–Savart Law

PYQ Priority: *****

Gives the magnetic field contribution $d\vec{B}$ produced by a small current element $Id\vec{l}$ at a distance r.

Biot–Savart Law

$$dB = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$$

Where $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T\cdot m/A}$ is the magnetic permeability of free space.

Biot–Savart Law (Vector Form)

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(d\vec{l} \times \hat{r})}{r^2} = \frac{\mu_0}{4\pi}\frac{I(d\vec{l} \times \vec{r})}{r^3}$$

2.7 Magnetic Field Due to an Infinite Straight Wire

PYQ Priority: *****

Calculates the magnetic field B at a perpendicular distance r from an infinitely long straight wire carrying a steady current I.

Field of Infinite Wire

$$B = \frac{\mu_0 I}{2\pi r}$$

Crucial Proportionally: $B \propto \frac{1}{r}$

2.8 Magnetic Field on the Axis of a Circular Coil

PYQ Priority: *****

Gives the magnetic field at a distance x along the axis of a circular current loop of radius R containing N turns.

Field on Axis of Circular Coil

$$B = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}$$

2.8.1 At the Centre of the Coil ($x = 0$):

Field at Centre of Circular Coil

$$B = \frac{\mu_0 N I}{2R}$$

2.9 Ampere's Circuital Law

PYQ Priority: *****

Statement: The line integral of the magnetic field $\vec{B}$ around any closed path (Amperian loop) is equal to μ_0 times the total net current passing through the surface enclosed by the loop.

Ampere's Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

2.10 Solenoid

PYQ Priority: *****

A long, tightly wound helical coil of wire designed to produce a uniform magnetic field.

Magnetic Field Inside Solenoid (Vacuum)

$$B = \mu_0 n I$$

Where $n = \frac{N}{l}$ is the number of turns per unit length.

Solenoid with Magnetic Core

$$B = \mu_0 \mu_r n I = \mu n I$$

Where μ_r is the relative permeability of the core material.

2.11 Toroid

PYQ Priority: ★★★★

A solenoid bent into a closed circular ring shape.

Magnetic Field of Toroid

$$B = \frac{\mu_0 N I}{2\pi r}$$

Where r is the mean radius of the toroid and N is the total number of turns.

2.12 Magnetization

The vector quantity representing the density of induced or permanent magnetic dipole moments in a magnetic material.

Magnetization Vector

$$\vec{M} = \frac{\text{Net Magnetic Dipole Moment}}{\text{Volume}}$$

Unit: A/m

2.13 Magnetic Susceptibility

A dimensionless proportionality constant indicating the degree of magnetization of a material in response to an applied magnetic field.

Magnetic Susceptibility

$$\vec{M} = \chi \vec{H}$$

Where $\vec{H}$ is the magnetic field intensity (auxiliary field).

2.14 Magnetic Permeability

Permeability Relations

$$\mu = \frac{B}{H}$$

Relative Permeability:

$$\mu_r = \frac{\mu}{\mu_0}$$

2.15 Relation Between Permeability and Susceptibility

PYQ Priority: ****

Derive the direct relationship between magnetic permeability and magnetic susceptibility.

Relation Between μ and χ

$$\mu = \mu_0(1 + \chi)$$

$$\mu_r = 1 + \chi$$

2.16 Classification of Magnetic Materials

2.16.1 Diamagnetic Materials

- Magnetization is opposite to the external field.
- **Susceptibility:** $\chi < 0$ (weakly negative).
- **Relative Permeability:** $\mu_r < 1$.
- Strongly repelled from high-field regions.
- **Examples:** Copper (Cu), Silver (Ag), Gold (Au), Water.

2.16.2 Paramagnetic Materials

PYQ Priority: ****

Materials that magnetize weakly in the direction of the external field.

- **Susceptibility:** $\chi > 0$ (weakly positive).
- **Relative Permeability:** $\mu_r > 1$.
- Weakly attracted to magnets.
- **Examples:** Aluminium (Al), Platinum (Pt), Oxygen (O_2).

2.16.3 Ferromagnetic Materials

- Strong permanent or induced magnetization in the field direction.
- **Susceptibility:** $\chi \gg 1$ (extremely large, positive).
- **Relative Permeability:** $\mu_r \gg 1$.

- Strongly attracted to magnets.
- **Examples:** Iron (Fe), Nickel (Ni), Cobalt (Co).

2.17 Comparison Table of Magnetic Materials

Table 1: Key Magnetic Properties Comparison

Property	**Diamagnetic**	**Paramagnetic**	**Ferromagnetic**
Susceptibility (χ)	Negative ($\chi < 0$)	Positive ($\chi > 0$)	Very Large Positive ($\chi \gg 1$)
Relative Permeability (μ_r)	$\mu_r < 1$	$\mu_r > 1$	$\mu_r \gg 1$
External Attraction	Weakly Repelled	Weakly Attracted	Strongly Attracted
Typical Examples	Cu, Ag, H_2O	Al, Pt, O_2	Fe, Ni, Co

Magnetostatics PYQ Priority: High-Yield Topics

Prioritize studying and writing out derivations for:

1. **Biot–Savart Law:** Integral and vector representation.
2. **Infinite Wire Field:** Derived via Biot–Savart and Ampere's Law.
3. **On-Axis Circular Coil Field:** Complete derivation.
4. **Ampere's Circuital Law:** Derivation and boundary conditions.
5. **Solenoid & Toroid:** Magnetic field formulas.
6. **Relation:** $\mu_r = 1 + \chi$.

Magnetostatics Revision Complete! ✓

3 Part 3: Electromagnetic Induction

3.1 Electromagnetic Induction (EMI)

The physical generation of an electromotive force (emf) across an electrical conductor in a changing magnetic field.

Core Concept of EMI

Changing Magnetic Flux (ϕ_B) $\implies$ Induced emf (e)

3.2 Magnetic Flux

Total magnetic field lines passing normally through a given area.

Magnetic Flux Formula

$$\phi_B = \vec{B} \cdot \vec{A} = BA\cos\theta$$

Where θ is the angle between the magnetic field $\vec{B}$ and the normal vector to the surface area $\vec{A}$.
Unit: Weber (Wb) = Tesla $\cdot$ m^2

3.2.1 Special Cases

- **Maximum Flux ($\theta = 0°$):** $\phi_B = BA$.
- **Zero Flux ($\theta = 90°$):** $\phi_B = 0$.

3.3 Faraday's First Law

PYQ Priority: *****

Statement: Whenever the magnetic flux linked with a closed circuit changes, an electromotive force (emf) is induced in the circuit, which lasts as long as the change in flux continues.

3.4 Faraday's Second Law

PYQ Priority: *****

Statement: The magnitude of the induced emf in a circuit is directly proportional to the time rate of change of magnetic flux linked with that circuit.

Faraday's Law of Induction

$$e = -\frac{d\phi_B}{dt}$$

For a Coil with N Turns:

$$e = -N\frac{d\phi_B}{dt}$$

Negative Sign: Indicates the direction of the induced emf (dictated by Lenz's Law).

3.5 Motional emf

Induced voltage generated across a conductor moving through a stationary magnetic field.

Motional emf

$$e = Blv$$

Where B is the magnetic field, l is the length of the conductor, and v is its perpendicular velocity.

3.6 Lenz's Law

PYQ Priority: *****

Statement: The direction of the induced current is always such that it opposes the change in magnetic flux that produced it.

3.6.1 Implications

- **Flux Increasing:** Induced current creates a magnetic field opposing the increase.
- **Flux Decreasing:** Induced current creates a magnetic field supporting the flux to prevent decrease.
- Directly explains the negative sign in $e = -\frac{d\phi_B}{dt}$.
- A direct consequence of the **Law of Conservation of Energy**.

3.7 Self-Induction

The production of an opposing induced emf in a coil when the current flowing through that same coil changes over time.

Self-Inductance

$$N\phi_B = LI \quad \Longrightarrow \quad L = \frac{N\phi_B}{I}$$

Where L is the self-inductance of the coil.
Unit: Henry (H) = Wb/A

Self-Induced emf

$$e = -L\frac{dI}{dt}$$

3.8 Self-Inductance of a Solenoid

PYQ Priority: *****

Derivation of self-inductance for a long solenoid of length l, area A, and total turns N.

Self-Inductance of Solenoid

$$L = \frac{\mu_0 N^2 A}{l}$$

3.8.1 Proportionality Trends

- $L \propto N^2$ (square of the number of turns).
- $L \propto A$ (cross-sectional area).
- $L \propto \frac{1}{l}$ (inversely proportional to length).

3.9 Self-Inductance of a Toroid

Self-Inductance of Toroid

$$L = \frac{\mu_0 N^2 r^2}{2R}$$

Where R is the mean radius and r is the cross-sectional radius.

3.10 Mutual Induction

The phenomenon of inducing an emf in a secondary coil when the current flowing in a primary coil changes.

Mutual Inductance

$$N_2\phi_{21} = MI_1 \quad \Longrightarrow \quad M = \frac{N_2\phi_{21}}{I_1}$$

Where M is the mutual inductance between the two coils.

Mutual Induced emf

$$e_2 = -M\frac{dI_1}{dt}$$

Mutual Inductance of Two Coaxial Solenoids

$$M = \frac{\mu_0 N_1 N_2 A}{l}$$

3.11 Current Density

Current flowing per unit cross-sectional area of a conductor.

Current Density

$$J = \frac{I}{A} \quad \Longrightarrow \quad I = \int \vec{J} \cdot d\vec{A}$$

Unit: $\mathrm{A/m^2}$

3.12 Continuity Equation

PYQ Priority: ⋆⋆⋆⋆ (Direct Derivation)

Expresses the global conservation of electric charge in differential form.

Continuity Equation

$$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

Where $\vec{J}$ is current density and ρ is charge density.

3.13 Displacement Current

PYQ Priority: ★★★★

The current that arises due to a time-varying electric field, introduced by Maxwell to resolve Ampere's law inconsistency.

Displacement Current

$$I_d = \epsilon_0 \frac{d\phi_E}{dt}$$

Displacement Current Density

$$J_d = \epsilon_0 \frac{\partial E}{\partial t}$$

3.14 Modified Ampere's Law (Ampere–Maxwell Law)

Ampere–Maxwell Law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}} + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

3.15 Maxwell's Equations

PYQ Priority: ★★★★★ (Crucial 10 Marks Question)

The four fundamental equations of classical electromagnetism, uniting electricity and magnetism.

3.15.1 Equation 1: Gauss's Law for Electricity

Gauss's Law (Electric)

Differential Form: $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

Integral Form: $\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$

3.15.2 Equation 2: Gauss's Law for Magnetism

Gauss's Law (Magnetic)

Differential Form: $\nabla \cdot \vec{B} = 0$

Integral Form: $\oint \vec{B} \cdot d\vec{A} = 0$

3.15.3 Equation 3: Faraday's Law of Induction

Faraday's Law

Differential Form: $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Integral Form: $\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$

3.15.4 Equation 4: Ampere–Maxwell Law

Ampere–Maxwell Law

Differential Form: $\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

Integral Form: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$

3.16 Maxwell's Equations Summary Table

Table 2: Summary of Maxwell's Equations

Equation Name	Differential Form	Integral Form
Gauss (Electric)	$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$	$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$
Gauss (Magnetic)	$\nabla \cdot \vec{B} = 0$	$\oint \vec{B} \cdot d\vec{A} = 0$
Faraday Law	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$
Ampere–Maxwell	$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$	$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$

Electromagnetic Induction PYQ Priority

Focus first on deriving and explaining:

1. **Maxwell's Equations:** Physical meaning of all 4 equations.
2. **Continuity Equation:** Detailed charge conservation derivation.
3. **Displacement Current:** Need for adding this term to Ampere's Law.
4. **Self-Inductance of a Solenoid:** Step-by-step derivation.
5. **Lenz's Law:** Verification of conservation of energy.

Electromagnetic Induction Revision Complete! ✓

4 Part 4: Network Theorems

4.1 Ohm's Law

Ohm's Law

$$V = IR$$

Where V is the voltage, I is the current, and R is the electrical resistance.

4.2 Electrical Power

Electrical Power Formulas

$$P = VI$$

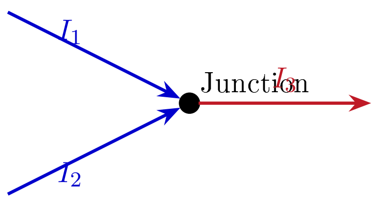
Alternative Forms:

$$P = I^2R \quad \text{and} \quad P = \frac{V^2}{R}$$

4.3 Kirchhoff's Current Law (KCL)

PYQ Priority: ★★★★

Statement: The algebraic sum of currents entering and leaving any node (junction) in an electrical network is zero.



Kirchhoff's Current Law

$$\sum I = 0 \implies I_{\text{in}} = I_{\text{out}}$$

In the diagram above: $I_1 + I_2 = I_3$.

Conservation Law: Based on the **Law of Conservation of Charge**.

4.4 Kirchhoff's Voltage Law (KVL)

PYQ Priority: ★★★★

Statement: The algebraic sum of all branch voltages and emf sources around any closed loop in a circuit is equal to zero.

Kirchhoff's Voltage Law

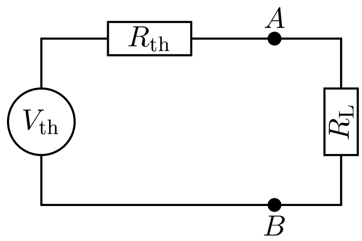
$$\sum V = 0$$

Conservation Law: Based on the **Law of Conservation of Energy**.

4.5 Thevenin's Theorem

PYQ Priority: ★★★★★ (Direct PYQ Proof)

Statement: Any linear, bilateral two-terminal network consisting of voltage/current sources and resistances can be replaced by an equivalent circuit containing a single voltage source V_{th} in series with an equivalent resistance R_{th}.



Thevenin Values

$$V_{\text{th}} = \text{Open-circuit voltage measured across terminals } AB$$
$$R_{\text{th}} = \text{Equivalent resistance looking back into } AB \text{ with active sources deactivated}$$

4.5.1 Procedure to Find Thevenin Equivalent

1. Remove the load resistance R_L from terminals A and B.
2. Calculate the open-circuit voltage V_{oc} across terminals A and B. This is V_{th}.
3. Deactivate all independent sources:
   - Replace independent voltage sources with a **short circuit**.
   - Replace independent current sources with an **open circuit**.
4. Calculate the equivalent resistance looking back into terminals A and B. This is R_{th}.
5. Reconnect R_L and compute the load current:

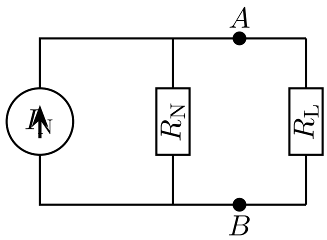
Thevenin Load Current

$$I_L = \frac{V_{\text{th}}}{R_{\text{th}} + R_L}$$

4.6 Norton's Theorem

PYQ Priority: ★★★★ (Direct PYQ Proof)

Statement: Any linear, bilateral active two-terminal network can be replaced by an equivalent circuit consisting of a single constant current source I_N in parallel with an equivalent resistance R_N.



Norton Values

$$I_N = \text{Short-circuit current } I_{sc} \text{ flowing between terminals } AB$$
$$R_N = \text{Thevenin resistance } R_{\text{th}}$$

Norton Load Current

$$I_L = I_N \left(\frac{R_N}{R_N + R_L} \right)$$

4.7 Thevenin–Norton Conversion

Source Transformations

Thevenin to Norton:

$$I_N = \frac{V_{\text{th}}}{R_{\text{th}}}$$

Norton to Thevenin:

$$V_{\text{th}} = I_N R_N$$

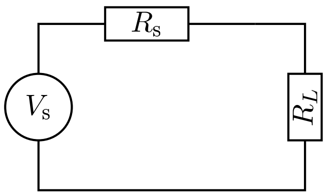
Resistances:

$$R_N = R_{\text{th}}$$

4.8 Maximum Power Transfer Theorem

PYQ Priority: *****

Statement: Maximum power is transferred from a source network to an adjustable load resistance R_L when the value of the load resistance is equal to the internal resistance R_s (or Thevenin equivalent resistance R_{th}) of the source network.



Theorem Condition

$$R_L = R_s \quad \text{or} \quad R_L = R_{\text{th}}$$

4.8.1 Derivation Framework

Power delivered to the load P_L:

$$P_L = I^2 R_L = \left(\frac{V_s}{R_s + R_L} \right)^2 R_L$$

To maximize power, set $\frac{dP_L}{dR_L} = 0$:

$$\frac{d}{dR_L} \left[\frac{V_s^2 R_L}{(R_s + R_L)^2} \right] = 0 \quad \Longrightarrow \quad R_L = R_s$$

Substituting $R_L = R_s$ yields the maximum power:

Maximum Power Formula

$$P_{\max} = \frac{V_s^2}{4R_s}$$

Efficiency at Max Power

$$\eta = 50\%$$

4.9 Theorem Comparison Table

Table 3: Key Network Theorem Summaries

Theorem	Equivalent Form	Key Relation / Output
Thevenin	Voltage source in series with R_{th}	$I_L = \frac{V_{\text{th}}}{R_{\text{th}}+R_L}$
Norton	Current source in parallel with R_N	$I_L = I_N \frac{R_N}{R_N+R_L}$
Max Power	Load matched to source resistance	$R_L = R_{\text{th}} \implies P_{\max} = \frac{V_{\text{th}}^2}{4R_{\text{th}}}$

Network Theory PYQ Priority

Focus first on the following topics:

1. **Thevenin's Theorem:** Step-by-step proofs and analytical circuit analysis.
2. **Maximum Power Transfer Theorem:** Proof of $R_L = R_{\text{th}}$, calculation of maximum power and 50% efficiency limit.
3. **Norton's Theorem:** Parallel equivalence and Norton–Thevenin transformation steps.
4. **Kirchhoff's Laws (KCL & KVL):** Conceptual base and physical conservation principles.

Network Theory Revision Complete! ✓

5 Entire Electricity & Magnetism Formula Sheet (Highest Priority)

This is a comprehensive reference sheet listing all crucial formulas across all four parts. Memorize these!

5.1 Electrostatics

- **Electric Field:** $E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$
- **Electric Potential:** $V = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}$
- **Potential Gradient Relation:** $E = -\frac{dV}{dr}$ or $\vec{E} = -\nabla V$
- **Dipole Moment:** $p = q(2a)$
- **Potential Due to Dipole:** $V_{\text{dipole}} = \frac{1}{4\pi\epsilon_0}\frac{p\cos\theta}{r^2}$
- **Axial Dipole Electric Field:** $E_{\text{axial}} = \frac{1}{4\pi\epsilon_0}\frac{2p}{r^3}$
- **Equatorial Dipole Electric Field:** $E_{\text{equatorial}} = \frac{1}{4\pi\epsilon_0}\frac{p}{r^3}$

5.2 Magnetostatics

- **Lorentz Force:** $F = qvB\sin\theta$
- **Force on Wire:** $F = BIL\sin\theta$
- **Biot–Savart Law:** $dB = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$
- **Infinite Straight Wire Field:** $B = \frac{\mu_0 I}{2\pi r}$
- **Circular Coil Axial Field:** $B = \frac{\mu_0 NIR^2}{2(R^2+x^2)^{3/2}}$
- **Solenoid Field:** $B = \mu_0 nI$

5.3 Electromagnetic Induction

- **Induced emf (Faraday's Law):** $e = -N\frac{d\phi_B}{dt}$
- **Self-Induced emf:** $e = -L\frac{dI}{dt}$
- **Self-Inductance of a Solenoid:** $L = \frac{\mu_0 N^2 A}{l}$
- **Mutual Inductance of Two Solenoids:** $M = \frac{\mu_0 N_1 N_2 A}{l}$
- **Continuity Equation (Charge Conservation):** $\nabla\cdot\vec{J} + \frac{\partial\rho}{\partial t} = 0$
- **Displacement Current:** $I_d = \epsilon_0\frac{d\phi_E}{dt}$

5.4 Network Theory

- **Ohm's Law:** $V = IR$
- **Thevenin Circuit Load Current:** $I_L = \frac{V_{\text{th}}}{R_{\text{th}}+R_L}$
- **Thevenin–Norton Conversion:** $I_N = \frac{V_{\text{th}}}{R_{\text{th}}}$ and $R_N = R_{\text{th}}$
- **Maximum Power Transfer Condition:** $R_L = R_{\text{th}}$
- **Maximum Power Output:** $P_{\max} = \frac{V_s^2}{4R_s}$

All E&M One-Shot Revision Content Successfully Transformed into LaTeX! ✓